

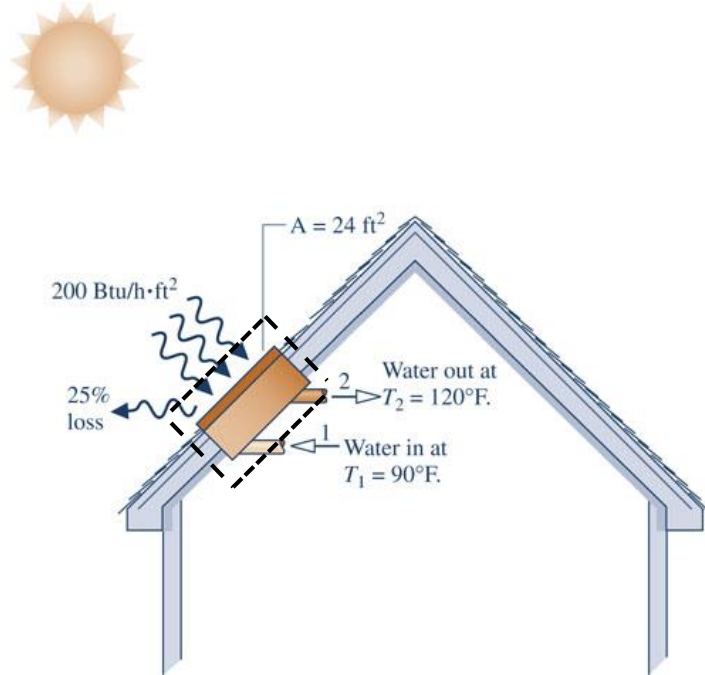
Problem 4.55

Figure P4.55 shows a solar panel embedded in a roof. The panel, which has a surface area of 24 ft², receives energy from the sun at a rate of 200 Btu/h per ft² of collector surface. Twenty-five percent of the incoming energy is lost to the surroundings. The remaining energy is used to heat domestic hot water from 90 to 120°F. The water passes through the collector with negligible pressure drop. Neglecting kinetic and potential energy effects, determine at steady state how many gallons of water at 120°F the collector generates per hour.

KNOWN: Data are provided for the steady state operation of a solar collector, producing hot domestic hot water.

FIND: Determine how many gallon of hot water the collector generates per hour.

SCHEMATIC AND GIVEN DATA:



ENGINEERING MODEL: (1) The control volume enclosing the solar collector is at steady state. (2) For the control volume, $\dot{W}_{cv} = 0$ and kinetic and potential energy effects can be neglected. (3) For liquid water, $v \approx v_f(T)$ and $h \approx h_f(T)$.

ANALYSIS:

Reducing Eq. 4.20a,

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} [h_1 - h_2 + \frac{v_1^2 - v_2^2}{2} + g(z_1 - z_2)]$$

$$\Rightarrow \dot{m} = \frac{\dot{Q}_{cv}}{h_2 - h_1} \quad (1)$$

where

$$\dot{Q}_{cv} = (0.75)(200 \frac{\text{Btu}}{\text{h} \cdot \text{ft}^2})(24 \text{ ft}^2) = 3600 \text{ Btu/h}$$

Then with $h_1 \approx h_f(90^\circ\text{F}) = 58.07 \text{ Btu/lb}$, $h_2 \approx h_f(120^\circ\text{F}) = 88.0 \text{ Btu/lb}$ from Table A-2E.

Inserting values in Eq. (1),

$$\dot{m} = \frac{3600 \text{ Btu/h}}{(88.0 - 58.07) \text{ Btu/lb}} = 120.3 \frac{\text{lb}}{\text{h}}$$

The volumetric flow rate of the water delivered at 120°F is then

$$\begin{aligned} (AV)_2 &= v_2 \dot{m} = v_f(120^\circ\text{F}) \dot{m} \\ &= (0.01621 \frac{\text{ft}^3}{\text{lb}}) (120.3 \frac{\text{lb}}{\text{h}}) \\ &= 1.95 \frac{\text{ft}^3}{\text{h}} \end{aligned}$$

Converting to gallons/h,

$$(AV)_2 = 1.95 \frac{\text{ft}^3}{\text{h}} \left| \frac{1 \text{ gal}}{0.13368 \text{ ft}^3} \right| = 14.59 \frac{\text{gal}}{\text{h}} \quad \leftarrow$$

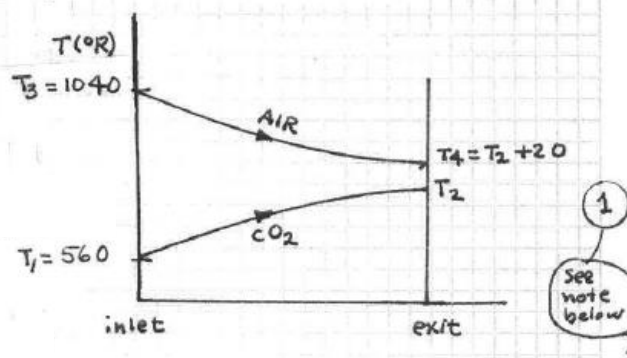
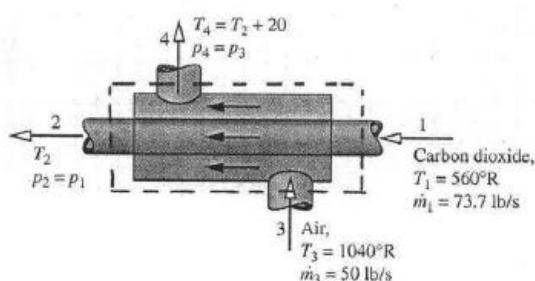
Problem 4.58

Figure P4.58 provides steady-state operating data for a *parallel flow* heat exchanger in which there are separate streams of air and carbon dioxide (CO₂). Stray heat transfer with the surroundings of the heat exchanger and kinetic and potential energy effects can be ignored. The ideal gas model applies to each gas. A constraint on the heat exchanger size requires the temperature of the exiting air to be 20 degrees greater than the temperature of the exiting CO₂. Determine the exit temperature of each stream, in °R.

KNOWN: State data are provided for a parallel flow heat exchanger operating at steady state with separate streams of air and carbon dioxide, each a gas.

FIND: Subject to the constraint that the temperature of the exiting air must be 20 degrees greater than the temperature of the exiting CO₂, determine the exit temperature of each stream, in °R.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL

1. The control volume shown in the schematic is at steady state.
2. For the control volume, $\dot{Q}_{cv}$ is negligible and kinetic and potential energy effects are ignored.
3. The ideal gas model applies to each gas.

ANALYSIS: Mass rate balances for the streams give $\dot{m}_2 = \dot{m}_1$ and $\dot{m}_4 = \dot{m}_3$.

An energy rate balance reduces as follows: $0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1(h_1 - h_2) + \dot{m}_3(h_3 - h_4)$

That is, $0 = \dot{m}_1(h_1 - h_2) + \dot{m}_3(h_3 - h_4)$, where $h_1 = h_{CO_2}(T_1)$ and $h_3 = h_{air}(T_3)$.

Also, $h_2 = h_{CO_2}(T_2)$ and $h_4 = h_{air}(T_2 + 20)$. T_2 can be determined iteratively

- ② using data from Tables A-22E and A-23E. A more direct approach is to use IT, as follows:

$T_1 = 560$ // °R
 $m_1 = 73.7$ // lb/s
 $T_3 = 1040$
 $m_3 = 50$

$$m_1(h_1 - h_2) + m_3(h_3 - h_4) = 0$$

$$T_4 = T_2 + 20$$

$$h_1 = h_T(\text{"CO2", } T_1)$$

$$h_2 = h_T(\text{"CO2", } T_2)$$

$$h_3 = h_T(\text{"Air", } T_3)$$

$$h_4 = h_T(\text{"Air", } T_4)$$

SOLUTION:

$$T_2 = 759.9 \text{ R}$$

$$T_4 = 779.9 \text{ R}$$

Problem 4.58 (Continued)

1. To meet a size constraint for a parallel flow heat exchanger can be challenging, for as the temperatures of the two streams approach each other the area required for heat transfer can become very large
2. An iterative solution using Tables A-22E and A-23E is left as an exercise.

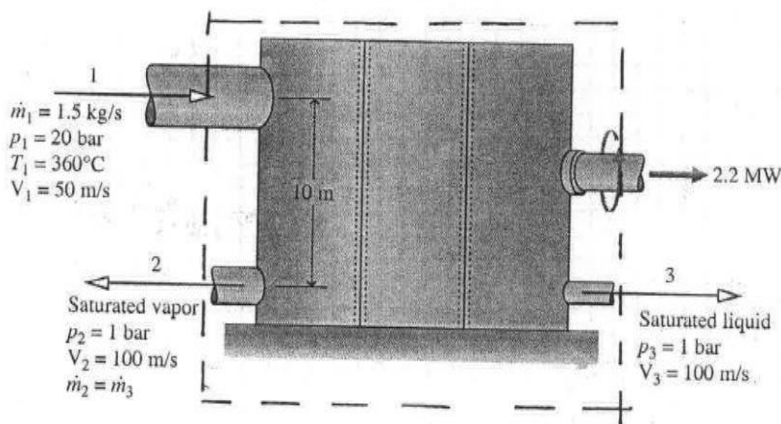
Problem 4.74

Figure P4.74 provides steady-state operating data for a cogeneration system with water vapor at 20 bar, 360°C entering at location 1. Power is developed by the system at a rate of 2.2 MW. Process steam leaves at location 2, and hot water for other process uses leaves at location 3. Evaluate the rate of heat transfer, in MW, between the system and its surroundings. Let $g = 9.81 \text{ m/s}^2$.

KNOWN: Steady-state data are provided for a cogeneration system operating with water as the working fluid.

FIND: Determine the rate of heat transfer between the system and its surroundings, in MW

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The control volume shown in the schematic is at steady state.
2. $g = 9.81 \text{ m/s}^2$

ANALYSIS: A mass rate balance reads, $\dot{m}_1 = \dot{m}_2 + \dot{m}_3$. Since $\dot{m}_2 = \dot{m}_3$, we get $\dot{m}_2 = \dot{m}_3 = \dot{m}_1/2$

An energy rate balance reduces to give this expression:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left[h_1 + \frac{V_1^2}{2} + gz_1 \right] - \dot{m}_2 \left[h_2 + \frac{V_2^2}{2} + gz_2 \right] - \dot{m}_3 \left[h_3 + \frac{V_3^2}{2} + gz_3 \right]$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left[h_1 - \frac{h_2}{2} - \frac{h_3}{2} \right] + \dot{m}_1 \left[\frac{V_1^2}{2} - \frac{V_2^2}{4} - \frac{V_3^2}{4} \right] + \dot{m}_1 g \left[z_1 - \frac{z_2}{2} - \frac{z_3}{2} \right]$$

where,

$$(i) \left[h_1 - \frac{h_2}{2} - \frac{h_3}{2} \right] = \left[3159.3 - \frac{2675.5}{2} - \frac{417.46}{2} \right] = 1612.82 \text{ kJ/kg} \quad (\text{Data from Tables A-3, 4})$$

$$(ii) \left[\frac{V_1^2}{2} - \frac{V_2^2}{4} - \frac{V_3^2}{4} \right] = \left[\frac{(50)^2}{2} - \frac{(100)^2}{4} - \frac{(100)^2}{4} \right] \left(\frac{\text{m}}{\text{s}} \right)^2 \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right| = -3.75 \frac{\text{kJ}}{\text{kg}}$$

$$(iii) g \left[z_1 - \frac{z_2}{2} - \frac{z_3}{2} \right] = (9.81 \frac{\text{m}}{\text{s}^2}) [10 \text{ m}] \left| \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right| = 0.1 \frac{\text{kJ}}{\text{kg}}$$

Collecting results.

$$0 = \dot{Q}_{cv} - 2.2 \text{ MW} + (1.5 \frac{\text{kg}}{\text{s}}) \left[1612.82 - 3.75 + 0.1 \right] \frac{\text{kJ}}{\text{kg}} \left| \frac{1 \text{ MW}}{10^3 \text{ kJ/s}} \right|$$

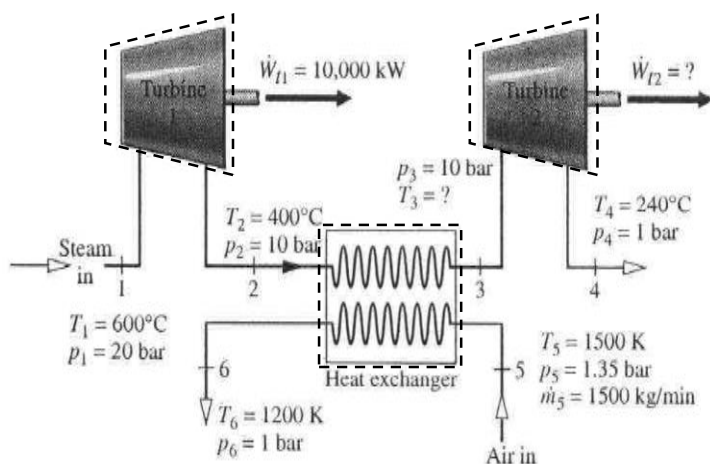
$$\textcircled{1} \Rightarrow \dot{Q}_{cv} = -0.21 \text{ MW} \leftarrow$$

1. The effects of kinetic and potential energy are minor in this application.

Problem 4.76

Separate streams of steam and air flow through the turbine and heat exchanger arrangement shown in Fig. P4.76. Steady-state operating data are provided on the figure. Heat transfer with the surroundings can be neglected, as can all kinetic and potential energy effects. Determine (a) T_3 , in K, and (b) the power output of the second turbine, in kW.

SCHEMATIC & GIVEN DATA:



ENGR. MODEL:

1. Consider a control volume at steady state enclosing each of the three components.
2. For each control volume, ignore stray heat transfer and kinetic and potential energy effects.
3. The ideal gas model applies for the air. (This can be verified from the compressibility chart.)

ANALYSIS:

- (a) To determine the steam mass flow rate write an energy rate balance for turbine 1 and use data from Table A-4:

$$\dot{W}_{t1} = \dot{m}_1 (h_1 - h_2) \Rightarrow \dot{m}_1 = \frac{\dot{W}_{t1}}{h_1 - h_2} = \frac{10,000 \text{ kW} \left| \frac{1 \text{ kJ/s}}{1 \text{ kW}} \right|}{(3690.1 - 3263.9) \text{ kJ/kg}}$$

$$= 23.46 \text{ kg/s}$$

Next, an energy rate balance for the heat exchanger reduces to

$$0 = \dot{m}_2 [h_2 - h_3] + \dot{m}_5 [h_5 - h_6]$$

with data from Table A-22,

$$\Rightarrow h_3 = h_2 + \frac{\dot{m}_5}{\dot{m}_2} [h_5 - h_6] = 3263.9 \frac{\text{kJ}}{\text{kg}} + \frac{(1500/60) \text{ kg/s} (1635.97 - 1277.79) \frac{\text{kJ}}{\text{kg}}}{23.46 \text{ kg/s}}$$

$$= 3645.6 \frac{\text{kJ}}{\text{kg}}$$

Interpolating in Table A-4 at 10 bar gives, $T_3 = 576^\circ\text{C} (849 \text{ K}) \leftarrow (a)$

- (b) An energy rate balance for turbine 2 is

$$\dot{W}_{t2} = \dot{m}_3 (h_3 - h_4) = 23.46 \frac{\text{kg}}{\text{s}} (3645.6 - 2954.5) \frac{\text{kJ}}{\text{kg}} \left| \frac{1 \text{ kW}}{1 \text{ kJ/s}} \right|$$

$$= 16,213 \text{ kW} \leftarrow (b)$$

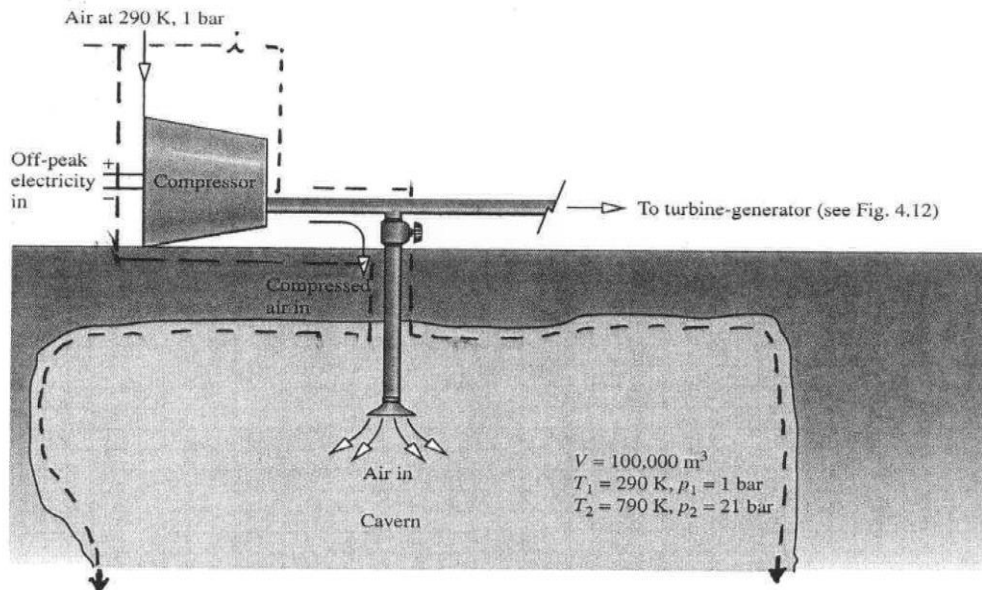
Problem 4.81

Figure P4.81 provides operating data for a compressed-air energy storage system using off-peak electricity to power a compressor that fills a cavern with pressurized air (See Sec.4.8.3). The cavern shown in the figure has a volume of 10^5 m^3 and initially holds air at 290 K, 1 bar, which corresponds to ambient air. After filling, the air in the cavern is at 790 K, 21 bar. Assuming ideal gas behavior for the air, determine (a) the initial and final mass of air in the cavern, each in kg, and (b) the work required by the compressor, in GJ. Ignore heat transfer and potential energy effects.

KNOWN: The compressor of a compressed-air energy storage system delivers air to a cavern. State data and cavern size are provided.

FIND: Determine the initial and final mass of air in the cavern, each in kg, and the work required by the compressor, in GJ.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The control volume is defined by the dashed line on the schematic. The condition at i remains constant at 290 K, 1 bar.
2. For the control volume, $\dot{Q}_{cv} = 0$ and kinetic and potential energy effects are ignored.
3. The air is modeled as an ideal gas.

ANALYSIS: For the one-inlet control volume, mass and energy rate balances reduce with listed assumptions to read, respectively, $\frac{dm_{cv}}{dt} = \dot{m}_i$,

$\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i h_i$. Combining these expressions and integrating,

$$\frac{dU_{cv}}{dt} = -\dot{W}_{cv} + h_i \frac{dm_{cv}}{dt} \Rightarrow \Delta U_{cv} = -W_{cv} + h_i [m_2 - m_1], \text{ which is a special case of Eq. 4.28 in Sec. 4.12.2.}$$

(1)

where

$$m_1 = \frac{p_1 V}{RT_1} = \frac{(10^5 \text{ N/m}^2)(10^5 \text{ m}^3)}{\left(\frac{8314 \text{ N}\cdot\text{m}}{28.97 \text{ kg}\cdot\text{K}}\right)(290 \text{ K})} = 1.2 \times 10^5 \text{ kg}$$

$$m_2 = \frac{p_2 V}{RT_2} = \frac{(21 \times 10^5 \text{ N/m}^2)(10^5 \text{ m}^3)}{\left(\frac{8314 \text{ N}\cdot\text{m}}{28.97 \text{ kg}\cdot\text{K}}\right)(790 \text{ K})} = 9.26 \times 10^5 \text{ kg}$$

Problem 4.81 (Continued)

with data from Table A-22,

$$u_1 = 206.91 \text{ kJ/kg}$$

$$u_2 = 584.21 \text{ kJ/kg}$$

$$h_1 = 290.16 \text{ kJ/kg}$$

Eq.(1) gives

$$\begin{aligned} (-W_{cv}) &= [(9.26 \times 10^5 \text{ kg})(584.21 \text{ kJ/kg}) - (1.2 \times 10^5 \text{ kg})(206.91 \frac{\text{kJ}}{\text{kg}})] \\ &\quad - (290.16 \frac{\text{kJ}}{\text{kg}})[9.26 \times 10^5 - 1.2 \times 10^5] \text{ kg} \end{aligned}$$

$$= [5.41 \times 10^8 \text{ kJ} - 0.25 \times 10^8 \text{ kJ}] - 2.34 \times 10^8 \text{ kJ}$$

$$= 2.82 \times 10^8 \text{ kJ} \left| \frac{1 \text{ GJ}}{10^6 \text{ kJ}} \right|$$

$$= 282 \text{ GJ}$$

4

Problem 4.86

A rigid tank having a volume of 0.1 m^3 initially contains water as a two-phase liquid-vapor mixture at 1 bar and a quality of 1 %. The water is heated in two stages:

Stage 1: Constant-volume heating until the pressure is 20 bar.

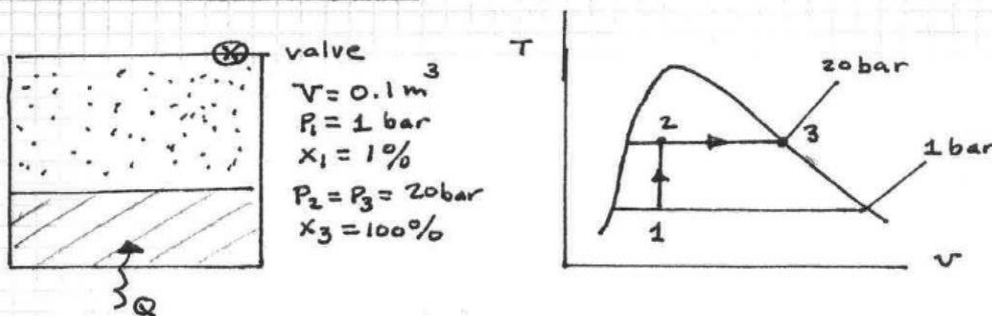
Stage 2: Continued heating while saturated water vapor is slowly withdrawn from the tank at a constant pressure of 20 bar. Heating ceases when all the water remaining in the tank is saturated vapor at 20 bar.

For the water, evaluate the heat transfer, in kJ, for each stage of heating. Ignore kinetic and potential energy effects.

KNOWN: A rigid tank, initially containing a two-phase liquid-vapor mixture, is heated in two stages: constant-volume heating followed by heating while saturated water vapor is withdrawn at a constant pressure. State data is provided.

FIND: For each stage of heating, evaluate the heat transfer, in kJ.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. For process 1-2, a closed system is considered that consists of the initial tank contents.
2. For process 2-3, a one-exit control volume is considered for which the condition at the exit is saturated vapor at 20 bar.
3. For each system, $\dot{W}_{cv} = 0$ and kinetic and potential energy effects are ignored.

ANALYSIS: We begin by finding key property values at states 1, 2, 3:

$$v_1 = v_f + x_1(v_g - v_f) = \frac{1.0432}{10^3} + 0.01 \left[1.694 - \frac{1.0432}{10^3} \right] = 0.018 \text{ m}^3/\text{kg}$$

$$v_2 = v_1 \Rightarrow x_2 = \frac{v_2 - v_f}{v_g - v_f} = \frac{0.018 - (1.1767/10^3)}{0.09963 - (1.1767/10^3)} = 0.17$$

$$m_1 = \frac{V}{v_1} = \frac{0.1 \text{ m}^3}{0.018 \text{ m}^3/\text{kg}} = 5.56 \text{ kg}, \quad m_2 = m_1$$

$$m_3 = \frac{V}{v_{g3}} = \frac{0.1 \text{ m}^3}{0.09963 \text{ m}^3/\text{kg}} = 1.0 \text{ kg}$$

$$u_1 = u_f + x_1[u_g - u_f] = 417.36 + 0.01[2506.1 - 417.36] = 438.25 \text{ kJ/kg}$$

$$u_2 = u_f + x_2[u_g - u_f] = 906.44 + 0.17[2600.3 - 906.44] = 1194.4 \text{ kJ/kg}$$

$$h_g = h_{g3} = 2799.5 \text{ kJ/kg}, \quad u_3 = u_{g3} = 2600.3 \text{ kJ/kg}$$

Problem 4.86 (Continued)

PROCESS 1-2:

$$\Delta U + \cancel{\Delta KE} + \cancel{\Delta PE} = Q - W \Rightarrow Q = m_1 [u_2 - u_1] \\ = 5.56 \text{ kg} [1194.4 - 438.25] \frac{\text{kJ}}{\text{kg}} \\ = 4204.2 \text{ kJ}$$

PROCESS 2-3

Mass and energy rate balances for the one-exit control volume reduce to, $\frac{dm_{cv}}{dt} = -\dot{m}_e$ and $\frac{dU_{cv}}{dt} = \dot{Q}_{cv} - \dot{m}_e h_e$

Combining them gives, $\frac{dU_{cv}}{dt} = \dot{Q}_{cv} + h_e \frac{dm_{cv}}{dt}$
↑
remains constant

Integrating with respect to time,

$$\Delta U_{cv} = Q_{cv} + h_e [m_3 - m_2] \Rightarrow (m_3 u_3 - m_2 u_2) = Q_{cv} + h_e [m_3 - m_2]$$

$$m_3 u_3 - m_2 u_2$$

Solving, $Q_{cv} = (m_3 u_3 - m_2 u_2) - h_e [m_3 - m_2]$, where $m_2 = m_1$.

Inserting values,

$$Q_{cv} = [(1 \text{ kg})(2600.3 \frac{\text{kJ}}{\text{kg}}) - (5.56 \text{ kg})(1194.4 \frac{\text{kJ}}{\text{kg}})] - (2799.5 \frac{\text{kJ}}{\text{kg}})[1 - 5.56] \text{ kg} \\ = 8725.2 \text{ kJ}$$

1. The total heat transfer is $(4204.2 \text{ kJ} + 8725.2 \text{ kJ}) = 12,929.4 \text{ kJ}$

An overall energy balance reads,

$$Q_{cv} = (m_3 u_3 - m_1 u_1) - h_e (m_3 - m_1) \\ = [(1 \text{ kg})(2600.3 \frac{\text{kJ}}{\text{kg}}) - (5.56 \text{ kg})(438.25 \frac{\text{kJ}}{\text{kg}})] - (2799.5 \frac{\text{kJ}}{\text{kg}})(1 - 5.56) \text{ kg} \\ = 12,929.4 \text{ kJ}$$

which checks the total given previously.